

Multidimensional Modelling in NoSQL Database

A Systematic Review

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Abstract — NoSQL Database Manager Systems (DBMS) are an alternative to analytical systems and have been used in this regard. As analytical systems often use data multidimensional modelling, the question arises of how to apply this modelling to NoSQL DBMS, which has such differentiated physical characteristics of relational databases. A systematic literature review (SLR) is performed to gather the aspects prompting the application of multidimensional modelling to NoSQL. One hundred twelve were pre-selected, and 67 were classified as relevant according to the research questions defined for this SLR. Three main approaches are highlighted: classical multidimensional modelling, multidimensional modelling with junctions of aggregated facts and dimensions in a single entity, and multidimensional modelling with aggregated facts and dimensions in a single entity, and adding pre-computed summaries. These approaches are explained, and their differences and similarities are detailed.

Keywords - NoSQL; multidimensional modelling; systematic literature review.

I. INTRODUCTION

Decision-making systems are fundamental to many organisations in the most diverse businesses. These systems are considered key parts of organisations business development, and for many years, they were seen as a competitive differential. In the 1990s, a set of these decision-making systems was defined as Business Intelligence (BI) and comprised modelling tools and technologies to analysis tools [1].

Data Warehouse (DW), a database dedicated to the decision-making context, is one of the main components in BI environments [2]. According to Lins and Ferreira [3], multidimensional modelling (or Dimensional Modelling or "Star Schema" Modelling) is an essential tool for building DW. Multidimensional modelling is intended to facilitate data recovery and analysis. DW and multidimensional modelling are the main facilitators for decision-making systems, especially as related to BI.

The main concepts of the multidimensional model are dimensions and facts. Dimensions contain segmentation, dimensionality, and granular analysis information. Facts store data associated with each dimension table. Star schema is generated by a fact table and multiple-dimension tables [4].

Both DW and multidimensional modelling are built based on a relational database (RDB) and use concepts of this persistence

system, namely, data tables related by foreign keys and referential integrity constraints. Multidimensional modelling is a normalised model regarding the 2nd Normal Form (2FN) [5] to reduce data redundancy and physical storage space [6].

3Vs mainly characterise Big Data: varied types, huge data volume, which increase at a colossal velocity, and with scaling-up accessible processing capacity according to the existing velocity processing requirement. The Big Data environment encompasses technology to process and analyse those data [7]. Similarly to BI, a Big Data environment generates information for decision support.

However, Relational Database Management Systems (RDBMS) are not ideal persistence structures for creating DWs for the current context of Big Data, mainly due to the difficulty in the scalability of such systems [8]. Therefore, distributed persistence models, such as NoSQL Database Management Systems (DBMS), are seen as better alternatives [8]–[12].

NoSQLs are non-relational DBMS that support large volumes of data with high performance and availability; they present their languages for data manipulation, facilitating data access. Other features include the absence of SQL query language, horizontal scalability and, in some cases, the lack of formal schema and eventual data consistency [13].

NoSQL DBMS present distinct characteristics when compared to traditional RDBMS, raising the following questions regarding the development of decision-making systems using NoSQL:

1. How has DW been built with NoSQL DBMS?
2. How has multidimensional data modelling been elaborated for NoSQL DBMS?
3. What are the differences between modelling and DW for different types of NoSQL?

In order to answer these questions, a Systematic Literature Review (SLR) is presented.

Ayed [14] presented a non-systematic literature review of multidimensional modelling and DW construction on NoSQL, but the study focuses solely on graph-oriented NoSQL.

So far, no other SLR on modelling and building DW in NoSQL has been found, considering the different kinds of NoSQL families. Thus, the proposed SLR contributes to the literature review. Unlike Ayed [14], it includes not only graph-

oriented but also other structures of NoSQL, such as column-oriented and document-oriented [13].

This paper is organised as follows. Section II introduces the procedure used to develop the SLR; its results are in Section III. Section IV details the main multidimensional modelling identified for the NoSQL. Finally, Section V concludes this paper.

II. PROCEDURE FOR THE SYSTEMATIC LITERATURE REVIEW

This SLR, or simply systematic review, adapted the model proposed by Kitchenham and Charters [15]. Figure 1 shows the steps defined for the SLR presented here. In addition, some activities were automated in this SLR, and programs written in Python language were created for this purpose. Some of the activities (associated with the selection phase) included in the review flow (Fig. 1) are part of this automation.

The planning stage (Fig. 1) establishes basic research questions, inclusion and exclusion criteria for the papers searched, selection of research databases, and the research evaluation criteria. The conducting stage comprises selecting and evaluating the papers searched, extracting data that characterises the article and synthesising these data. In SLR, inclusion/exclusion criteria were applied twice: the first one, for pre-selection, by an automated process; the second one, the selection of the pre-selected papers, was manual.

Once the main research questions were defined (Fig. 1), the following points were established:

- Language: in which the papers must be expressed: only papers written in English.
- Period: from January 2012 to December 2022 were selected.
- Publishing methods: papers peer-review from scientific congresses, and journals were considered.
- Selection of research databases: the selected databases were ACM Digital Library, IEEE Explorer, Scopus (Elsevier), Springer and Web of Science.
- Search strings: the search tag was defined to be executed in all the databases mentioned, regardless of the resources offered by each. The "tag" was written to cover as many papers as possible addressing the main subjects of the three research questions: NoSQL, Data Warehouse and Multidimensional Modelling. Thus, the tag was: "'data warehouse' OR 'olap' OR 'star schema' OR 'multidimensional' AND 'NoSQL'".

As part of the automated process, a set of papers were selected. These pre-selected papers went through another manual selection step (step "Apply manual exclusion criteria" in Fig. 1), in which each had its title and abstract read. Papers that fit the previously defined exclusion criteria were excluded from the list of papers that would be read entirely. There were three previously described exclusion criteria:

- Non-peer reviewed papers

- The paper quotes NoSQL but does not present concepts applicable to NoSQL
- The paper does not address any aspect of logical modelling for the data warehouse in NoSQL

As a result of the manual selection step, a final set of papers was selected for a full reading. These were read and classified according to the criteria previously defined.

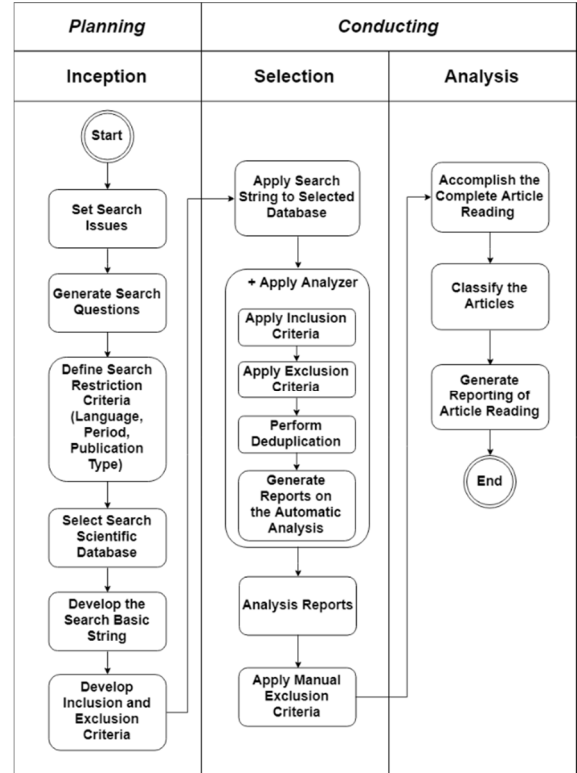


Figure 1. Workflow followed for SLR

III. PROCEDURE FOR THE SYSTEMATIC LITERATURE REVIEW

The results were divided into the selection and reading of papers. As a result, papers were classified into classical normalised multidimensional modelling (CNMM), fully non-normalised multidimensional modelling (FNMM), and fully non-normalised multidimensional modelling with pre-computed aggregate fields (FNMMPAF).

A. Results of Selection of Papers

Table 1 shows the distribution of papers recovered from the research databases according to their type of publication (journal or conference proceedings). The papers by the research databases are distributed only to show the results; it is not the objective or scope of this work to classify them. Eight thousand two hundred seventy-one papers were initially searched.

TABLE I. NUMBER OF PAPERS PER RESEARCH DATABASE CLASSIFIED BY THEIR TYPE OF PUBLICATION

Database	Journals	Conferences	Publications total
ACM	164	360	524
IEEE	36	329	365
Science Direct	797	0	797
Scopus	495	561	1,056
Springer	450	197	647
Web of Science	1,949	2,933	4,882
Total	3,891	4,380	8,271

After selecting and excluding papers, 112 (one hundred twelve) were selected for full reading, approximately 1.35% of the initial number of papers. Since retrieving the entire document of [16] and [17] was impossible, these references were excluded from the final list of papers for reading.

About 51% of the pre-selected papers were found in multiple databases, and 34.75% were excluded from the manual selection process. We noted that from the 112 articles selected, 75 were published in congresses and conferences. This shows the current relevance of the topic in the research since the publication process in journals generally takes longer than in conferences.

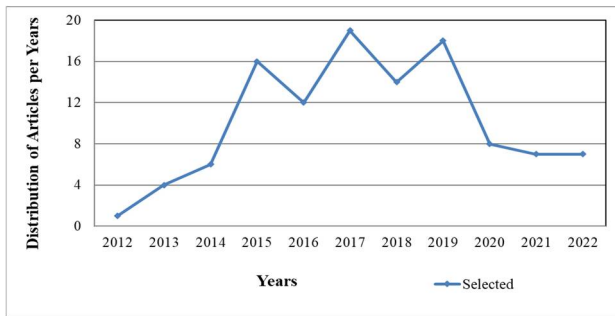


Figure 2. Distribution of Papers per Year in the Selection Step

Figure 2 shows the distribution of papers from 2012 to 2019; themes related to NoSQL and multidimensional modelling have increased during this period. From 2020 to 2022, there was a vast reduction in the publication on this theme.

B. Results of Reading of Papers

Only papers that treated multidimensional modelling for NoSQL were considered relevant. Thus, after reading 112 papers, 67 (~60%) were classified as suitable.

According to Figure 3, column and document-oriented families are the most widely used (approximately 77% of the cases). Document-oriented is the NoSQL most popular, achieving 44% of the papers cited; among this type of NoSQL, MongoDB is the most used. NoSQL Columns Family appears in the second position in most research, and Cassandra and HBase are the principal NoSQL mentioned in this family. Several works employ more than one NoSQL family, besides different NoSQL implementations.

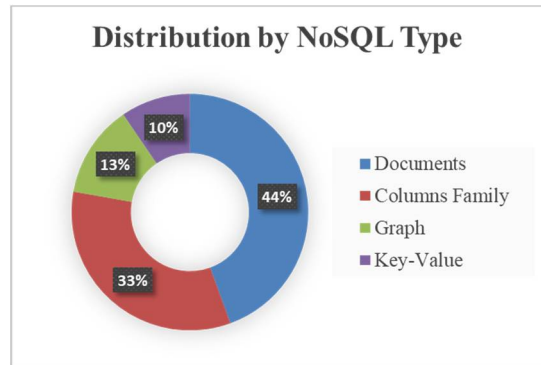


Figure 3. Distribution by NoSQL Type of Cited in Reading Articles

Most articles selected for this SLR had models based on the star schema model [4]. More about the data model is explained in section IV. As a result, this SLR evidenced a substantial difference between the multidimensional modelling for RDB and NoSQL and how these models are mapped for an analytical model.

IV. DISCUSSION – MULTIDIMENSIONAL MODELLING IN NoSQL

Three major types of multidimensional modelling in NoSQL stand out in SLR, which we named Classical Normalised Multidimensional Modelling (CNMM), Fully Non-Normalised Multidimensional Modelling (FNMM) and Fully Non-normalised Multidimensional Modelling with Pre-computed Aggregate Fields (FNMMPAF). These models are based on the traditional star schema modelling and are adapted according to the physical model characteristic of each type of NoSQL.

A. Classical Normalised Multidimensional Modelling

CNMM represents the mapping of classic multidimensional modelling to the database, then CNMM is strongly indicated for graph-oriented NoSQL since this family allows representing relationships. Figure 4 shows the CNMM related to the star schema for graph-oriented NoSQL.

The classical modelling proposed by Kimball [4] was also applied to other NoSQL families. However, as other NoSQL families have problems dealing with relationships, the application of this modelling was criticised. Other NoSQL families, such as the document-oriented and column families, are not able to solve entity joining problems by themselves once these NoSQL do not work with relationships. To perform the join operations, these NoSQL need additional tools.

NoSQL-oriented graphs have internal structures specific to join operations and high-performing aggregations [16]. Azzini [17] presented the OLAP proposal on NoSQL. A more detailed explanation of CNMM over Graph NoSQL can be seen in [18], which introduced the "Graph Cube" model developed by Zhao et al. [19] (Figure 4).

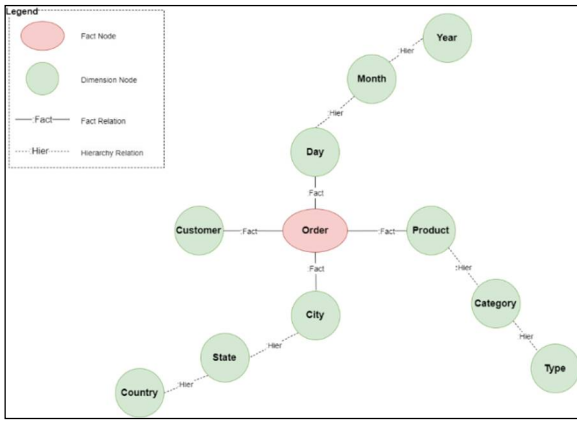


Figure 4. Graph-oriented NoSQL Multidimensional Modelling [18]

Graph Cube represents data cubes built by graphs, in which dimensions are described in node dimensions; facts are based on counts represented by a fact node or in edges connecting nodes of dimensions and the fact node. Figure 4 shows the link between the fact and dimension nodes defined by edges with the tag Fact and the dimension hierarchy represented by edges with the tag Hier. The main articles describing CNMM are in [16]–[25].

B. Fully Non-Normalised Multidimensional Modelling

FNMM is a fully non-normalised multidimensional model with a high redundancy but better query performance. Concerning FNMM, we observed that it was employed by all the papers selected. This approach is applied to all NoSQL families, except for the graph-oriented family, since this one focuses on relationships of entities and has better adherence to classical multidimensional modelling or CNMM. FNMM presents different nomenclatures in the literature, such as Flat Model [27], WideTable [28], StarTable, and others.

Chevalier et al. [25] propose a physical structure for FMNN whereby facts and dimensions belong to the same document; however, each fact and dimension is represented as a non-nested subdocument. Pereira et al. [29] employ the complete denormalisation of the schema within the documents, using a document for each fact and its dimensions (Figure 5). They besides presented FNMM built-in as nested documents in their proposal.

Another option of physical modelling for column-oriented NoSQL is proposed in [30], outlined in Figure 6, in which facts and dimensions belong to the same table; however, facts and each dimension are represented in separated column families. The main articles defining FMNN can be seen in [24], [25], [27]–[54].

C. Fully Non-Normalised Multidimensional Modelling With Pre-computed Aggregate Fields

In FNMMPAF, the initial computation of data that needs to be summarised is also performed in addition to gathering facts and dimensions. This approach is based on Multidimensional On-Line Analytical Processing (MOLAP), in which data are pre-processed, and data aggregations are calculated in advance.

The MOLAP concept is employed by several analytical systems [19].

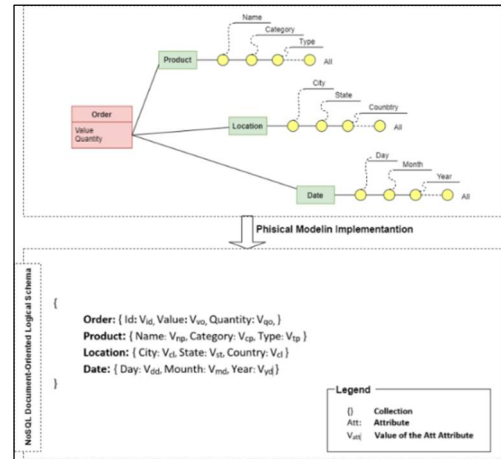


Figure 5. Physical Implementation of Modelling on Non-Nested document-oriented NoSQL [26]

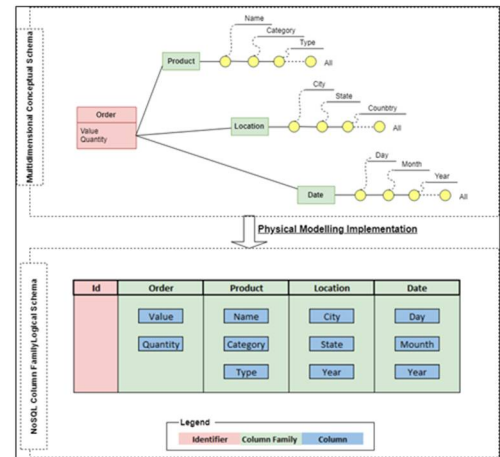


Figure 6. Physical Modelling Implementation in Column-Family Oriented NoSQL [26]

For document-oriented NoSQL, Chevalier et al. [52] and Stantic [55] address issues of summarisation, hierarchy ordering, and indexes, emphasizing that the required dimensions of documents should be pre-computed. Franciscus et al. [56] named FNMMPAF as Data Cube and Carniel et al. [24] as materialised view (MV), and they emphasise that both document and column-family-oriented NoSQL should maintain groups previously processed and results of aggregation functions. Zhao and Ye [19] and Chevalier et al. [45] named FNMMPAF a cuboid.

The previous aggregation is employed in classical modelling since it is an approach used for relational database analytical systems [25]. Therefore, this approach is applied to classical multidimensional modelling for graph-oriented NoSQL [18]. The main articles considered to define FNMMPAF can be seen in [19], [24], [45], [48], [52], [55]–[62].

V. CONCLUSION

NoSQL is used as an important Database Management System for transactional systems due to its high scalability, availability and processing power. Besides, due to these characteristics, NoSQL has been employed in multidimensional modelling for analytical systems.

SLR searches for publications in several scientific databases to identify state of the art for applications of multidimensional modelling using NoSQL families. From an initial total of 8,271 articles, following the procedure adopted for the proposed SLR, 112 articles were selected for a complete reading. As a result of a full reading, 67 articles were classified as relevant, according to the research questions defined.

Three types of multidimensional modelling were concluded to be mapped in NoSQL: a classic form of multidimensional modelling (CNMM), a fully denormalised form of multidimensional modelling (FNMM) and a fully denormalised form of multidimensional modelling with pre-computed aggregation (FNMPAF). Thus, CNMM is the adaptation of Star Schema logical modelling to the NoSQL physical model. The CNMM is implemented for the graph-oriented NoSQL family because it allows implementing relationships. FNMM is the total denormalisation of CNMM applied to NoSQL. Total denormalisation is performed when all data entities, facts and dimensions are listed in a single data entity, document, or table. This, joined into a single data entity, can generate data duplication. As well as FNMPAF being built on the FNMM concept, it added pre-computed data from aggregate fields. The result is thus denormalised data into a single data entity with the addition of pre-computed aggregate data. Therefore, FNMPAF is very similar to MOLAP models for relational databases. Consequently, these points justify the significant use of FNMM compared to the others and its application to the market and other organisations.

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